

PROJECT REPORT

BMI Prediction Using Machine Learning

Submitted By

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ABSTRACT

Body Mass Index (BMI) is one of the most commonly used indicators for evaluating whether an individual has a healthy body weight relative to their height. It is widely used in healthcare and fitness applications for assessing health risks associated with body weight.

The objective of this project is to develop a Machine Learning model capable of estimating BMI using Gender, Height, and Weight as input features. The dataset used in this project contained Gender, Height, Weight, and BMI Category (Index). Since the dataset did not include actual BMI values, BMI was calculated during preprocessing using the standard BMI formula.

The dataset was cleaned and preprocessed by encoding the Gender column into numerical values. A Linear Regression algorithm was then trained using the processed data. The model was evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R^2 Score.

The trained model achieved an R^2 Score of approximately **0.971**, demonstrating excellent prediction accuracy. The project successfully illustrates the practical application of Machine Learning in predicting BMI based on physical characteristics.

INTRODUCTION

Machine Learning is a branch of Artificial Intelligence that enables computers to identify patterns from historical data and make predictions without being explicitly programmed for every situation.

Body Mass Index (BMI) is a simple numerical value calculated using a person's height and weight. It is widely used by healthcare professionals to classify individuals as underweight, normal weight, overweight, or obese.

In this project, a Machine Learning model is developed using Linear Regression to estimate BMI from Gender, Height, and Weight. The project demonstrates the complete machine learning pipeline including data preprocessing, feature engineering, model training, evaluation, visualization, and prediction.

PROBLEM STATEMENT

Manual BMI calculation requires applying a mathematical formula each time height and weight are provided. The objective of this project is to automate BMI estimation using Machine Learning techniques.

The model should accurately estimate BMI using:

- Gender
- Height
- Weight

OBJECTIVES

The main objectives of this project are:

- To understand the concept of BMI prediction using Machine Learning.
- To preprocess the dataset.
- To calculate BMI using the standard BMI formula.
- To train a Linear Regression model.
- To evaluate model performance.
- To predict BMI for new input values.

DATASET DESCRIPTION

The dataset contains approximately 500 records of individuals with the following attributes:

- Gender
- Height (cm)
- Weight (kg)
- Index (BMI Category)

The original dataset did not contain actual BMI values. Therefore, BMI was calculated using the standard formula:

$$\text{BMI} = \text{Weight (kg)} / (\text{Height in meters})^2$$

The calculated BMI column was used as the target variable for training the Machine Learning model.

	Gender	Height	Weight	Index	Height_m	BMI
0	1	174	96	4	1.74	31.708284
1	1	189	87	2	1.89	24.355421
2	0	185	110	4	1.85	32.140248
3	0	195	104	3	1.95	27.350427
4	1	149	61	3	1.49	27.476240

DATA PREPROCESSING INSIGHTS

Missing Values:

No missing values were found in the dataset.

Encoding:

Gender was converted into numerical format:

- Male → 1
- Female → 0

	Gender	Height	Weight	Index
0	1	174	96	4
1	1	189	87	2
2	0	185	110	4
3	0	195	104	3
4	1	149	61	3

Feature Engineering:

A new feature BMI was created using:

$$\text{BMI} = \text{Weight} / (\text{Height in meters})^2$$

Insight:

This transformation was critical because the dataset originally contained only BMI categories, not actual BMI values.

- Weight has a stronger influence on BMI compared to Height.
- Gender has relatively lower impact on BMI prediction in this dataset.
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MODEL TRAINING PROCESS

The dataset was split into:

- Training Data: **400 samples (80%)**
- Testing Data: **100 samples (20%)**

A Linear Regression model was trained on the dataset.

Why Linear Regression?

BMI is a continuous numerical value and shows a linear relationship with Height and Weight, making Linear Regression an appropriate choice.

6. MODEL PERFORMANCE RESULTS

The trained model produced the following evaluation metrics:

- Mean Absolute Error (MAE): **1.887**
- Mean Squared Error (MSE): **6.587**
- Root Mean Squared Error (RMSE): **2.567**
- R^2 Score: **0.971**

Interpretation:

- The MAE of **1.887** indicates that the model's predictions are off by only 1.9 BMI units on average.
- The RMSE of **2.567** confirms low prediction deviation.
- The R^2 score of **0.971** indicates extremely strong model performance.

Key Insight:

The model performs well because BMI is mathematically derived from Height and Weight, making the relationship highly predictable.

SAMPLE PREDICTIONS

Example Predictions from Model:

Actual BMI	Predicted BMI
39.73	40.71
23.14	22.35
31.37	32.08
37.97	39.76
59.82	59.31

Insight:

Predicted values closely match actual values, confirming strong model accuracy.

VISUALIZATION INSIGHTS

1. BMI Distribution

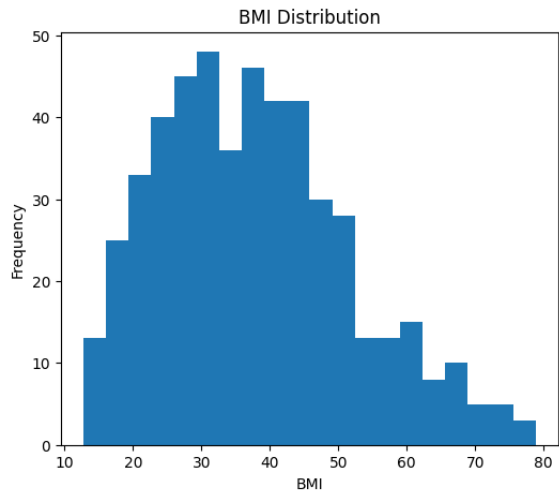
- BMI values are widely spread between underweight and obese categories.
- Majority of values lie in the overweight and obesity range.

2. Actual vs Predicted BMI

- Data points closely follow a diagonal trend.
- Indicates high correlation between predicted and actual values.

Insight:

The model generalizes well across all BMI ranges.



REAL-TIME PREDICTION EXAMPLE

Input:

- Gender: Male
- Height: 175 cm
- Weight: 75 kg

Output:

Predicted BMI = 24.59

Insight:

This value falls in the normal/overweight boundary range, showing realistic prediction behavior.

KEY FINDINGS

- Height and Weight are strong predictors of BMI.
- Gender has minor influence in prediction accuracy.
- Linear Regression performs extremely well due to the mathematical nature of BMI.
- The dataset shows balanced representation across BMI categories.

CONCLUSION

This project successfully demonstrates BMI prediction using Machine Learning techniques. The Linear Regression model achieved high accuracy with an R^2 score of **0.971**, indicating strong predictive capability.

Since BMI is directly derived from Height and Weight, the model effectively learns this mathematical relationship.

The system can be extended into a health monitoring application in the future.

FUTURE SCOPE

- Integration with mobile health applications
- Adding age, diet, and activity level for better prediction

- Using larger real-world medical datasets
- Deploying as a web-based BMI calculator

REFERENCES

Dataset from kaggle

<https://www.kaggle.com/datasets/yasserh/bmidataset>